

NEXORA

PHASE 1 PROCUREMENT TEST PLAN

Document ID: NEXORA-TP-PROC-001 | Version 1.1 | 21 August 2026

*ISO/IEC/IEEE 29119-aligned, risk-based test documentation
for the Procurement foundation before automation and AI integration*

Document control	Value
Owner	Pranidhi Peiris — NEXORA Quality Engineering
Status	Baseline draft for review and execution
Test level	System, integration, API, UI and end-to-end
Automation	Playwright + TypeScript (after manual stability)
Applicable release	NEXORA Phase 1 — Procurement

1. Purpose and test objectives

This test plan defines the governance, scope, approach, evidence, entry/exit criteria and controls for NEXORA Phase 1 Procurement testing. It is intended to establish a reliable, traceable quality baseline before large-scale Playwright automation and the later RAG, LLM and LangGraph layers.

- Verify that the complete procurement chain performs the intended business process and enforces valid status transitions.
- Detect functional, integration, data-integrity, authorisation, concurrency and auditability defects before automation is treated as a trusted signal.
- Maintain bidirectional traceability from requirement to test case, execution result, defect and automation test.
- Produce repeatable evidence suitable for later AI-assisted coverage and failure analysis.

2. Standards alignment and tailoring

This pack is tailored to the test-process concepts of ISO/IEC/IEEE 29119-2:2021 and the test-documentation templates of ISO/IEC/IEEE 29119-3:2021. IEEE 829 is not used as the governing standard because it has been superseded by the ISO/IEC/IEEE 29119 series. Alignment means that relevant content elements and controls are represented; it does not constitute independent certification or a claim of full organisational conformance.

Reference	Use in this pack
ISO/IEC/IEEE 29119-2:2021	Risk-based test planning, monitoring/control, design, implementation, execution and completion activities.
ISO/IEC/IEEE 29119-3:2021	Test plan, test case/specification, traceability, test results, incident/defect and completion-report information.
ISO/IEC/IEEE 29119-1:2022	Terminology and conceptual consistency.
IEEE 829-2008	Historical reference only; superseded by ISO/IEC/IEEE 29119.

3. Test items and Phase 1 scope

3.1 In scope

Test item	Coverage
Supplier Management	List, create, view, edit, activate/deactivate, validation, duplicate prevention and downstream inactive-supplier restriction.
Warehouse Management	Warehouse creation/view/selection, active-state eligibility and duplicate/invalid-data controls needed by procurement.
Purchase Requests	Create, view, edit, submit, cancel, line validation, status transitions, search and source data persistence.
Approval Workflow	Approval queue, approve/reject, comments, authorisation, segregation of duties, stale/duplicate actions and approval history.
Purchase Orders	Creation from approved requests, calculations, edit, cancellation, reapproval on material change, source linkage and duplicate prevention.
Goods Receipt	Full and partial receipt, remaining quantity, warehouse selection, over-receipt prevention, audit data and duplicate prevention.
Inventory Integration	Correct item/location balance update, exactly-once posting and linked inventory transaction history.
Security Controls	Authentication/session validity, role and object authorisation, malicious input, CORS/CSRF, sensitive-data exposure, rate limits, mass assignment, audit integrity, headers and supply-chain scanning.
Cross-cutting controls	Authentication/authorisation, API/UI error handling, transaction integrity, status history, auditability and safe retry behaviour.
End-to-end flow	Supplier → Purchase Request → Approval → Purchase Order → Goods Receipt → Inventory Update, including negative rejection/cancellation flow.

3.2 Out of scope for Phase 1

- Supplier invoice capture, three-way matching, accounts payable, bank/payment processing and financial posting.
- Sales, manufacturing, finance, HR, logistics and business-intelligence modules beyond procurement dependencies.

- Production-scale performance/load/stress testing, penetration testing and formal security certification; only essential functional security controls are included.
- AI/RAG/LLM/LangGraph implementation, autonomous agents and AI-generated decisions.
- Mobile-native applications, third-party supplier portals, EDI and external banking integrations unless later added to the release.
- Data migration, disaster recovery and production deployment validation unless separately commissioned.

4. Assumptions, dependencies and constraints

- A dedicated non-production PostgreSQL database and test environment are available and may be reset or seeded safely.
- Business rules such as required fields, uniqueness, permitted statuses, approval thresholds and reapproval triggers are approved or explicitly recorded as open decisions.
- At least two role-separated test users are available: requester/administrator and authorised approver.
- Supplier, item and warehouse master data can be created without using real personal or commercial information.
- Dates, currency, tax and rounding rules are configured consistently for the Phase 1 environment.
- The current system is a modular monolith; API, service and database effects can be observed during testing.

5. Risk-based test strategy

Risk	Impact	Priority control
Unauthorised or self approval	Unapproved spending and segregation-of-duty breach	Role/API negative tests; audit verification
Duplicate submission	Duplicate supplier, PO, receipt or inventory movement	Double-click, retry and replay tests; DB uniqueness
Invalid status transition	Documents bypass approval or become unusable	State-transition tests at UI and API layers
PO changed after approval	Spend exceeds approved value	Reapproval and approval-history tests
Over/duplicate receipt	Inventory and liability misstatement	Boundary, remaining-quantity and idempotency tests
Wrong warehouse/item update	Incorrect stock availability	Before/after DB reconciliation by item/location
Partial transaction commit	Orphan documents or inconsistent stock	Failure simulation and rollback/recovery verification
Weak error handling	Sensitive data disclosure or poor diagnosis	Safe API/UI error-content checks

5.1 Detailed risk register

Risk ID	Risk	L	I	Score	Level	Primary treatment	Residual	Owner
R-001	Unauthorised/self approval	3	5	15	Critical	Role, API, object and segregation tests	Low	Quality Engineer
R-002	Duplicate document or inventory posting	4	5	20	Critical	Idempotency, uniqueness, replay and DB checks	Medium	Developer
R-003	Invalid status transition	3	5	15	Critical	State model and server-side transition enforcement	Low	Developer
R-004	Approved PO changed without reapproval	4	5	20	Critical	Material-change rule and immutable history tests	Low	Product Owner
R-005	Over/duplicate goods receipt	4	5	20	Critical	Remaining quantity, concurrency and exactly-once tests	Medium	Developer
R-006	Incorrect item/warehouse stock update	3	5	15	Critical	Item-location reconciliation and transaction linkage	Low	Quality Engineer
R-007	Partial transaction commit	3	5	15	Critical	Rollback, failure simulation and recovery validation	Medium	Developer
R-008	Broken object/role authorisation	3	5	15	Critical	Negative API access tests and audit review	Low	Technical Owner
R-009	Injection or unsafe rendered input	3	5	15	Critical	Server validation, parameterisation and output encoding	Low	Developer
R-010	Sensitive data/secret exposure	3	5	15	Critical	Response/log review, secret scan and redaction	Low	Technical Owner
R-011	Performance degradation under concurrency	3	4	12	High	Baseline timing now; dedicated k6 phase later	Medium	Technical Owner
R-012	Incomplete or misleading test	3	4	12	High	Permanent IDs, evidence standard and RTM	Low	Quality

Risk ID	Risk	L	I	Score	Level	Primary treatment	Residual	Owner
	evidence					control		Engineer

Scoring model: Likelihood (L) and Impact (I) use 1–5; score = L × I. Suggested levels: Critical 15–25, High 10–14, Medium 5–9, Low 1–4. Residual risk must be reviewed after execution and before release approval.

6. Test approach

6.1 Test levels and techniques

Area	Approach
Manual functional testing	Execute documented happy paths, negative paths, field validations and state transitions before automating a scenario.
API testing	Verify methods, payloads, status codes, validation, authorisation, conflict handling and response contracts.
Database verification	Confirm uniqueness, relationships, status/audit records, transaction integrity and exactly-once inventory effects.
UI automation	Use Playwright for stable critical user journeys, visible validation, navigation and role behaviour.
API automation	Use Playwright APIRequestContext for deterministic setup, contract assertions and faster business-rule coverage.
End-to-end testing	Run one controlled golden path and one rejection/cancellation path across all Phase 1 components.
Techniques	Equivalence partitioning, boundary-value analysis, decision tables, state-transition testing, error guessing and risk-based prioritisation.

6.2 Execution sequence

1. Confirm requirement and expected rule.
2. Prepare isolated test data.
3. Execute manually across UI/API/DB as applicable.
4. Record actual result and evidence.
5. Log, fix and retest defects.
6. Automate stable Critical and High scenarios.
7. Run end-to-end regression and publish completion report.

6.3 Automation selection criteria

- Stable behaviour and approved expected result.
- Critical/high business risk or frequent regression value.
- Repeatable setup and deterministic assertion.
- No dependency on uncontrolled external systems.
- Test data can be generated, isolated and cleaned up.

6.4 Phase 1 security test specification

Security testing is risk-based and supports functional security assurance for the Phase 1 release. It does not replace an independent penetration test or formal security certification. The detailed SEC test cases cover:

- Authentication, invalid/expired sessions and logout behaviour.
- Role-based and object-level authorisation, including direct API bypass and privilege escalation attempts.
- SQL-injection-style input, stored/reflected script payloads and unsafe mass assignment.
- CORS and CSRF controls according to the selected authentication design.
- Sensitive information in responses and permitted application logs.
- Rate limiting/abuse protection for authentication and sensitive actions.
- Approval/status audit integrity and ordinary-user tamper prevention.
- HTTPS/security-header baseline, dependency vulnerability scanning and repository secret scanning.

Any Critical security finding blocks Phase 1 exit until corrected or formally risk-accepted by the authorised release owner. Credentials, attack payload evidence and sensitive logs must be stored only in approved restricted locations.

7. Test environment and test data

Component	Baseline
Frontend	NEXORA React application in a controlled test environment
Backend	FastAPI procurement APIs
Database	PostgreSQL test database
Automation	Playwright with TypeScript; HTML/JSON reporting; trace and screenshot on failure
Browsers	Chromium baseline; add cross-browser coverage when UI stabilises
Evidence	Test ID, timestamp, environment/build, expected/actual, API response, screenshot/trace and DB query where required

Test data must use synthetic supplier/contact information. Credentials, tokens, session cookies, secrets and production data must never be stored in test documents, repositories or AI knowledge sources.

8. Entry, suspension, resumption and exit criteria

8.1 Entry criteria

- Phase 1 feature is deployed to the test environment and smoke-accessible.
- Relevant requirement/status rules are available and unresolved decisions are recorded.
- Test users, roles, master data and database access are available.
- Known blockers are documented and the build is identifiable.

8.2 Suspension and resumption

Suspend affected testing for unavailable environments, corrupt/shared test data, repeated blocking failures, or a build that cannot complete the basic smoke path. Resume after the root condition is corrected, impacted data is reset, the build is identified, and the smoke test passes.

8.3 Exit criteria

- 100% of planned Critical test cases executed.
- All Critical cases pass; no open Critical defects.
- At least 95% of High cases pass, with any exception risk-accepted and documented.
- No unresolved defect permits unauthorised approval, duplicate financial/stock effect, invalid status bypass or incorrect inventory posting.
- Critical regression cases are automated and passing in the agreed baseline browser/API suite.
- E2E-001 and E2E-002 pass with traceable evidence.
- Requirements traceability has no unexplained Critical coverage gaps.
- Test completion report is reviewed and approved.

9. Test deliverables and configuration control

ID / artefact	Purpose
NEXORA-TP-PROC-001	Phase 1 Procurement Test Plan (this document)
NEXORA-TC-PROC-001	Detailed test cases and scenarios workbook
NEXORA-RTM-PROC-001	Requirement-to-test traceability matrix within workbook
NEXORA-TER-PROC-001	Test execution register within workbook
NEXORA-DL-PROC-001	Defect log within workbook
NEXORA-TCR-PROC-001	Test completion report to be produced at exit
Playwright report	Automated execution evidence using the same permanent test IDs

10. Roles and responsibilities

Role	Responsibilities
Quality Engineer / Test Lead	Plan, design, execute, automate, report, maintain evidence and recommend release readiness.
Developer	Resolve defects, support observability, review technical risks and provide deployable builds.
Business/Product Owner	Approve business rules, priorities, exceptions, status transitions and acceptance decisions.
Approver/Test User	Execute or support role-specific validation without sharing credentials.
Release decision owner	Accept residual risk and approve Phase 1 completion/deployment.

11. Defect management and reporting

Every failed test is evaluated before being classified as an application defect. The record includes Test ID, build/environment, reproducible steps, expected/actual result, severity, priority, evidence, affected requirement, owner and retest result. Suggested severity model: Critical—security/data-integrity/business-flow stop; High—major function broken with no acceptable workaround; Medium—limited function or workaround exists; Low—minor usability/cosmetic issue.

Progress reporting will include planned/executed/pass/fail/blocked counts, requirement coverage, Critical/High open defects, automation status, known risks and exit-criteria status. Counts in the workbook are formula-driven from editable records.

11.1 Coverage definitions

Coverage measure	Definition	Interpretation
Requirement design coverage	Requirements with one or more mapped tests ÷ total requirements	Shows planned traceability; does not show execution or quality.
Execution coverage	Executed tests ÷ planned tests	Executed excludes Not Run; report Not Applicable separately.
Pass rate	Passed tests ÷ executed tests	Report only when at least one test has executed.
Critical pass rate	Passed Critical tests ÷ executed Critical tests	Primary release-risk indicator.
Automation coverage	Automated tests ÷ automation-eligible planned tests	Not equivalent to code coverage.
Requirement pass coverage	Requirements whose mapped executed tests all pass ÷ total requirements	Shows evidenced requirement acceptance.
Security execution coverage	Executed SEC tests ÷ planned SEC tests	Phase 1 functional security progress.
Defect closure rate	Closed defects ÷ total logged defects	Must be read with severity and reopen rate.

11.2 KPI and release dashboard

The workbook dashboard calculates planned cases, Critical cases, requirements, mapped requirements, execution coverage, pass/fail/blocked rates, Critical pass rate, automation coverage, security execution, open defects by severity, closure rate, retest status and a release-readiness indicator. KPI values are decision support only; release approval remains a human responsibility.

KPI	Target / decision rule
Requirement design coverage	100% for approved Critical requirements; gaps explained for all others.
Critical execution coverage	100%.
Critical pass rate	100%.
Overall High-priority pass rate	≥95%, with documented exception acceptance.
Open Critical defects	0.
Open High defects	0 unless formally risk-accepted.
E2E-001 and E2E-002	Both Pass.
Security Critical cases	100% executed and passed.
Release status	READY only when mandatory exit rules are satisfied; otherwise NOT READY.

12. Requirements coverage baseline

Requirement ID	Area	Requirement summary	Risk
SUP-REQ-001	Supplier	Create a supplier with a unique supplier code and valid mandatory data	Critical
SUP-REQ-002	Supplier	Validate mandatory fields, formats, boundaries and whitespace	High
SUP-REQ-003	Supplier	Prevent duplicate supplier records according to configured uniqueness rules	Critical

Requirement ID	Area	Requirement summary	Risk
SUP-REQ-004	Supplier	View, search and edit supplier details with persisted changes	High
SUP-REQ-005	Supplier	Activate or deactivate a supplier and enforce downstream restrictions	Critical
WH-REQ-001	Warehouse	Create, view and select an active warehouse	High
WH-REQ-002	Warehouse	Prevent duplicate warehouse codes and invalid warehouse data	High
PR-REQ-001	Purchase Request	Create and save a purchase request with valid header and line data	Critical
PR-REQ-002	Purchase Request	Validate mandatory fields, quantities and request line completeness	Critical
PR-REQ-003	Purchase Request	View and edit a purchase request while its status permits editing	High
PR-REQ-004	Purchase Request	Submit a valid draft purchase request exactly once	Critical
PR-REQ-005	Purchase Request	Cancel an eligible purchase request and prevent invalid reuse	High
PR-REQ-006	Purchase Request	Enforce the approved purchase-request status transition model	Critical
APR-REQ-001	Approval	Present submitted requests to an authorised approver	Critical
APR-REQ-002	Approval	Approve or reject an eligible request and require an appropriate comment where configured	Critical
APR-REQ-003	Approval	Enforce approver permissions and segregation of duties	Critical
APR-REQ-004	Approval	Maintain immutable approval actor, timestamp, decision and comments history	High
APR-REQ-005	Approval	Prevent duplicate or stale approval actions	Critical
PO-REQ-001	Purchase Order	Create a purchase order only from an eligible approved request	Critical
PO-REQ-002	Purchase Order	Calculate line totals and order total accurately	Critical
PO-REQ-003	Purchase Order	View and edit a purchase order while rules permit	High
PO-REQ-004	Purchase Order	Require reapproval when an approved purchase order is materially changed	Critical
PO-REQ-005	Purchase Order	Cancel an eligible purchase order and block prohibited downstream activity	High
PO-REQ-006	Purchase Order	Prevent duplicate purchase-order creation and preserve source linkage	Critical
GRN-REQ-001	Goods Receipt	Receive eligible purchase-order quantities into the selected warehouse	Critical
GRN-REQ-002	Goods Receipt	Support full and partial receipt while maintaining remaining quantity	Critical
GRN-REQ-003	Goods Receipt	Prevent over-receipt, zero/negative quantities and duplicate receipt submission	Critical
GRN-REQ-004	Goods Receipt	Record receipt number, user, timestamp, PO and warehouse references	High
INV-REQ-001	Inventory	Increase on-hand inventory exactly once after a valid goods receipt	Critical
INV-REQ-002	Inventory	Update the correct item and warehouse without affecting unrelated stock	Critical
INV-REQ-003	Inventory	Maintain an auditable inventory transaction linked to the goods receipt	High
SEC-REQ-001	Security	Enforce authentication and session/token validity for protected procurement resources	Critical
SEC-REQ-002	Security	Enforce object-level and role-based authorisation for every procurement action	Critical
SEC-REQ-003	Security	Validate and safely handle malicious or unexpected input	Critical
SEC-REQ-004	Security	Protect browser-originated state-changing requests and apply secure cross-origin controls	High
SEC-REQ-005	Security	Prevent credential, token, personal and commercially sensitive data exposure	Critical
SEC-REQ-006	Security	Apply rate limiting and abuse protection to authentication and sensitive actions	High
SEC-REQ-007	Security	Maintain tamper-resistant security and approval audit events	Critical
SEC-REQ-008	Security	Prevent unsafe mass assignment of restricted fields	Critical
SEC-REQ-009	Security	Use appropriate transport, browser and API security controls	High
SEC-REQ-010	Security	Manage dependencies, secrets and application logs without known critical exposure	High
SYS-REQ-001	Cross-cutting	Return meaningful API and UI errors without exposing sensitive information	High
SYS-REQ-002	Cross-cutting	Record document status history consistently across procurement documents	High
SYS-REQ-003	Cross-cutting	Prevent duplicate records during retry, double-click and concurrent submission	Critical
SYS-REQ-004	Cross-cutting	Enforce authenticated role-based access to procurement actions	Critical
E2E-REQ-001	End-to-End	Complete Supplier to Inventory procurement flow with traceable document linkage	Critical

13. Approval

Role	Name	Decision	Date
Quality Engineer / Test Lead	Pranidhi Peiris	Pending	
Product / Business Owner		Pending	
Technical Reviewer		Pending	

Appendix A — Authoritative references

- ISO/IEC/IEEE 29119-2:2021, Software and systems engineering — Software testing — Part 2: Test processes. <https://www.iso.org/standard/79428.html>
- ISO/IEC/IEEE 29119-3:2021, Software and systems engineering — Software testing — Part 3: Test documentation. <https://www.iso.org/standard/79429.html>
- IEEE 829-2008 status page (superseded by the ISO/IEC/IEEE 29119 series). <https://standards.ieee.org/standard/829-2008.html>